Life-Long Learning With Continual Spectral-Spatial Feature Distillation for Hyperspectral Image Classification

Wenzhi Zhao, *Member, IEEE*, Rui Peng, Qiao Wang, Changxiu Cheng, William J. Emery, *Life Fellow, IEEE*, and Liqiang Zhang, *Member, IEEE*

***Abstract*— The rapid development of hyperspectral remote sensing technology, has led to an explosion in the number of available hyperspectral images (HSI). The fast and accurate characterization of HSI poses a significant challenge for remote sensing scientists. Currently, deep learning strategies with various neural networks have been successfully applied for HSI classification using the concept of the "dataset-model." Still, there is a need to develop universal deep learning models for HSI classification using a continual updating strategy. This article presents a life-long learning strategy to continually update model weights with the help of continual spectral-spatial feature distillation. Specifically, the proposed method introduces a spectral-spatial distillation strategy to retain knowledge of the previous well-trained model. Meanwhile, the learning metric term is integrated into a multilevel feature extraction to minimize the spectral-spatial feature discrepancy between the previous model and the new one. The experimental results indicate that our method achieves superior performance for continual HSI classification tasks without suffering from the persistent loss of characterization memory.**

***Index Terms*— Deep learning, hyperspectral image classification, knowledge distillation (KD), life-long learning, remote sensing.**

I. Introduction

HYPERSPECTRAL imagery collected by high-resolution remote sensing sensors are characterized by hundreds of narrow spectral bands. Each pixel in hyperspectral images (HSI) represents a spectral curve that captures the reflectance property of the target. The accurate classification of HSI is important in both remote sensing research and applications such as land cover identification, environment monitoring and mineral resource prospecting [1], [2], [3].

To differentiate various objects in HSI, many diverse classification methods have been proposed. Early classification methods include spectral-based curve matching such as the spectral angle mapper, and spectral information divergence (SID). Because of the high dimensionality of spectral profiles, however, it is difficult to find efficient spectral feature representations for HSI classification. Instead of using the full spectrum for HSI classification, dimension reduction methods reduce the number of spectral bands. Both supervised and unsupervised strategies have been proposed for dimension reduction, such as principal component analysis (PCA) [4], nonparametric weighted feature extraction (NWFE) [5], and local discriminant embedding (LDE) [6]. Althoughs the low-dimension representation features can be extracted, it is still difficult to discriminate the different objects having similar spectral properties.

Spatial features also play an important role in HSI classification. Recently, intensive studies have focused on combined spectral-spatial feature extraction. For example, hand-crafted geometric features, extended morphological features (EMPs), and attribute profiles have been applied in HSI spectral-spatial classification. Differently, deep learning methods can automatically extract high-level representative features that have hierarchical structures. To fully exploit spatial-spectral features of HSI, numerous modified deep learning networks have been proposed, including convolutional neural networks (CNN), ResNet, and generative adversarial networks.

The current deep learning frameworks for efficient HSI spectral-spatial feature extraction are, however, mainly focused on training and testing using the same dataset under the assumption of independent and identical distribution (i.i.d.). The rapid development of hyperspectral remote sensing technology has greatly increased the number of available HSIsimages acquired by a variety of cutting-edge satellites (e.g., Zhuhai, Gaofen-5) available for classification analysis. Thus, today image analysts are faced with the diverse suite of spectral/spatial resolutions of various hyperspectral remote sensors, which greatly increases the challenge of HSI classification

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Wenzhi Zhao and Rui Peng are with the State Key Laboratory of Remote Sensing Science, Institute of Remote Sensing Science and Engineering, Faculty of Geographical Science, Beijing Normal University, Beijing 100875, P. R. China, and also with the Beijing Engineering Research Center for Global Land Remote Sensing Products, Institute of Remote Sensing Science and Engineering, Faculty of Geographical Science, Beijing Normal University, Beijing 100875, P. R. China.

Qiao Wang and Liqiang Zhang are with the Faculty of Geographical Science, State Key Laboratory of Remote Sensing Science, Beijing Normal University, Beijing 100875, China.

Changxiu Cheng is with the State Key Laboratory of Earth Surface Processes and Resource Ecology, Beijing Normal University, Beijing 100875, China, and also with the Faculty of Geographical Science, Beijing Normal University, Beijing 100875, China (e-mail: chengcx@bnu.edu.cn).

William J. Emery is with the Colorado Center for Astrodynamics Research, University of Colorado, Boulder, CO 80309 USA.

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with intraclass variation and interclass similarities in the spectral-spatial domain.

Faced with the challenge of a shift from one data source of HSI to another, the pretrained deep learning networks often fail to accurately characterize the new dataset that has significant differences. To facilitate the cross-domain classification of HSI, a method of transfer learning is applied in HSI classification. In order to adapt the old knowledge to fit a new dataset, transfer learning tries to modify previous weights to fit the new dataset. There are two main approaches for transfer learning in the field of HSI classification [7]: (1) fine-tuning that retrains the previous (source domain) model using a few selected samples from the new (target) domain, and (2) data distribution adaptation, that maps a common feature space for both source and target domain to reach a similar distribution [8]. For the first approach, Yang et al. [9] propose a fine-tuning method to extract spectral-spatial features from the well-trained two-channel model. Subsequently, improved fine-tuning methods are used for HSI classification [10]. For the second method, Wang et al. [11] introduce deep domain adaptation to find a common selection for efficient HSI classification for both old and new HSI data sets.

Although the transfer learning method minimizes spectral-spatial feature discrepancies between the source and target domains, there is an increase in transfer times (more than two times) for a specific well-trained network. It inevitably forgets previously learned class-specific spectral-spatial features (source and target are not independent identical distribution), which is called the problem of "catastrophic forgetting." Specifically, with consecutive fine-tuning or domain adaptation operations, the inherited weights will gradually shift from the previous domain to a new parameter space in order to satisfy the new HSI classification task. Inevitably, the final well-trained model failed to retain previous knowledge, which constrains the universal ability in HSI classification with a huge expense of computational cost. How to preserve old knowledge on spectral-spatial feature extraction and include new knowledge on new datasets is the main challenge for practical applications in the HSI classification realm.

In order to handle the problem of "catastrophic forgetting" in a constant transfer learning process, continual learning [12] is proposed to retain previous knowledge inside a deep network without suffering from significant forgetting. Continual learning forces deep neural networks to have the ability to continuously learn new data distributions while maintaining their performance on previous tasks. In the scope of continual learning, there are three groups of knowledge preservation strategies. The first group is "experience replay-based methods," which build on and store a memory of the knowledge learned from previous classification tasks and preserve them for future utilization. For example, fixed memory can store the most representative samples for each class [13]. In addition, dynamic episodic memory of parameter gradients can also maintain the old knowledge in the continuous learning process [14]. But, the replay of prior knowledge usually requires additional parameter space to store previous data distributions, which is inefficient in terms of high-level feature extraction for complex remote sensing imagery. For the second group, regularization terms are applied to constrain the update process of the deep network weights. To serve this purpose, elastic weight consolidation (EWC) uses a Fisher information matrix to measure the importance of gradient updates for the new tasks [15]. Also, an Orthogonal Gradient Descent (OGD) retains the previous knowledge by adding additional projections from subspaces of new tasks [16]. Regularization-based methods may, however, gradually forget old knowledge without a proper knowledge distillation (KD) operation, especially with various remote sensing images. Last, the third group is called the parameter isolation/expansion method, which introduces a gating term to control subsets of the model that are dedicated to specific tasks [17], [18], [19], [20], [21]. For example, Packnet [22] frees up redundant parameters after the training of each task through consecutive pruning. At the same time, to ensure the learning capacity of deep networks, the parameter expansion technique is proposed to meet this requirement. To handle new tasks, dynamically expandable networks (DENs) and dynamically expanded networks are proposed to include new knowledge consecutively [18]. Still, the parameter isolation/expansion strategy faces significant challenges from a bulk load of sub-models and parameters. Therefore, the direct application of the above strategies for HSI classification on different datasets is not feasible.

Because of the differences in terms of spectral bands, spatial resolution and other sensor characteristics, the spectral-spatial feature representation shifts across different HSI datasets. Two problems strongly hinder the accurate classification of HSI across different satellite platforms: (1) how to transfer useful spectral-spatial feature representations from previous well-trained models to new ones? Currently, KD strategies have been proposed to extract effective knowledge in a "teacher-student" fashion [23]. The direct transfer of spectral-spatial information, however, remains unsolved. (2) How to maintain robustness in their classification? Because of the intra-class variation and inter-class similarity in HSI imagery, it is difficult to extract generic feature representations for efficient classification. The minimization of spectral-spatial features shifting between the previous model and the new one needs to be solved.

Therefore, in order to achieve the purpose of efficient HSI classification by continually implying previous knowledge, we propose a deep metric continual learning (DMLL) scheme with spectral-spatial feature distillation. Specifically, the DMLL first introduced a spectral-spatial distillation strategy to retain knowledge of the previous well-trained model. Subsequently, the metric learning term is integrated with multilevel feature extraction to minimize the spectral-spatial feature discrepancy between the earlier model and the new one. The contributions of DMLL are: 1) the proposed two-branch structure integrated spatial and spectral information provides robust feature representations across different HIS datasets; and 2) the feature-response KD scheme enables the model to preserve historical knowledge effectively for continual HIS classification tasks.

The rest of this article is organized as follows. Section II introduces the related previous studies. Section III introduces the method proposed in this article in detail. Section IV

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introduces the data sets, then introduces the experimental setup, and finally analyzes and discusses the experimental results. The last section presents the conclusions of this article.

II. Previous Related Studies

In this section, we provide an overview of the recent research on KD, which aims at knowledge conservation and low-computational cost requirements Then, we discuss applications of metric learning methods in terms of effective feature representation exploration as applied to remote sensing image classification.

A. Knowledge Distillation

To design knowledge preserving and computational cost-effective deep models, KD is proposed to distill the knowledge from a larger model to a lightweight smaller network. Since activation, neurons and intermediate features all contain different knowledge, there are three different categories of knowledge during KD process according to their different forms [24]: response-based knowledge, feature-based knowledge, and relation-based knowledge. Over the past decades, response-based knowledge has been widely used in transferring knowledge from a deep "teacher network" to a smaller "student network." The primary purpose is to minimize the divergence between the final prediction of the student model and the teacher model. Hinton et al. [25] first, matched the logits of teacher model and student model, which effectively distillate informative dark knowledge from teacher model to student model. It has been widely employed for model compress [26] and domain adaption [27]. Although response-based knowledge is usually applied to different tasks because of the simplicity and effectiveness, it ignores intermediate-level features for student model training. To further acquiring valuable knowledge from intermediate layers, many researchers have developed feature-based KD. For example, Romero et al. [28] introduced intermediate representations from the teacher model's hidden layer, which effectively improves the performance of student model. Komodakis and Zagoruyko [29] transfer intermediate feature knowledge by mimicking the attention map of the teacher model. Finally, relation-based knowledge further exploits the relationships among different layers or samples. Park et al. [30] present a relational knowledge distillation (RKD) to transfer the mutual relationship of data samples into the student model however, there is still a difficulty in modeling the relationship information from intermediate features or data samples as knowledge.

If we consider the different update strategies, the existing KD schemes can be categorized into offline, online distillations and self-distillation. The offline distillation pays more attention to knowledge transference rather than teacher model training. Typically, the whole distillation process consists of two steps, first pre-training the teacher model and subsequently guiding the student model training through logits or intermediate features KD. For instance, Hinton et al. [25] designed logits distillation loss, which is utilized to distill the knowledge of the pre-trained teacher model into the new model. Asif et al. [31] proposed an EKD model for KD from pre-trained teacher model, which consists of several CNN models into their compact student models; however, pre-training the high-capacity teacher model is still time-consuming, especially distillation of knowledge into a lightweight model. Consequently, online distillation overcomes the limitation of offline distillation by updating the teacher model and student model simultaneously. For instance, Zhang et al. [32] proposed deep mutual learning strategies that train multiple networks simultaneously, each network can be considered as either a teacher model or a student model, allowing the acquired model to outperform the static teacher model. Lan et al. [33] presented the On-the-fly Native Ensemble (ONE) KD strategy for learning in both small and large models, effectively improving the generalization of the trained model and reducing the consumption of computing resources significantly. To further improve the training efficiency and quality, Anil et al. [34] proposed a co-distillation strategy, which trained several similar architecture models with the same dataset in parallel. As for self-distillation, the same networks are adopted by the teacher model and the student model. Specifically, Zhang et al. [35] divides the model into multiple sections, and then transfers knowledge from the deeper sections to the shallower sections. To sum up, it is necessary to design an appropriate KD strategy by simultaneously considering response time and mid-level feature constrains for the continual updated HSI classification tasks.

B. Metric Learning

HSI classification using metric learning is one of the hottest topics in recent years. The main idea of metric learning is to decrease the distance between intra-class samples and further increase the distance between inter-class samples, which results in similar objects being aggregated and dissimilar objects staying separate in the latent space. Traditional metric learning, however, usually relied on the measurements of Euclidean space, or Mahalanobis, Matusita, Bhattacharyya, and Kullback–Leibler distances, where Mahalanobis distances enable data to be transformed into a more discriminative feature space. For example, Pasolli et al. [36] used Mahalanobis distance to measure the distance between two samples in order to exploit valuable labeled information in HSI. The approach based on these metrics, however, tends to focus on linear transformations rather than dealing with nonlinear knowledge and without kernel functions. Since concepts of deep learning were also proposed such as deep metric learning with activation function, which leads to better performance due to the advantage of the use of a hierarchical neural network.

Metric learning based on deep neural networks has been dramatically developed recently. Deep metric learning is concerned with projecting semantically similar samples to a closer place in the feature space [37]. To keep similar samples closer together and dissimilar samples of different categories apart in the feature space, numerous methods extract supervised information from unlabeled samples through defined different loss functions. Most of the existing deep metric learning methods can be categorized into three areas according to their sampling strategy and metric loss function: contrastive

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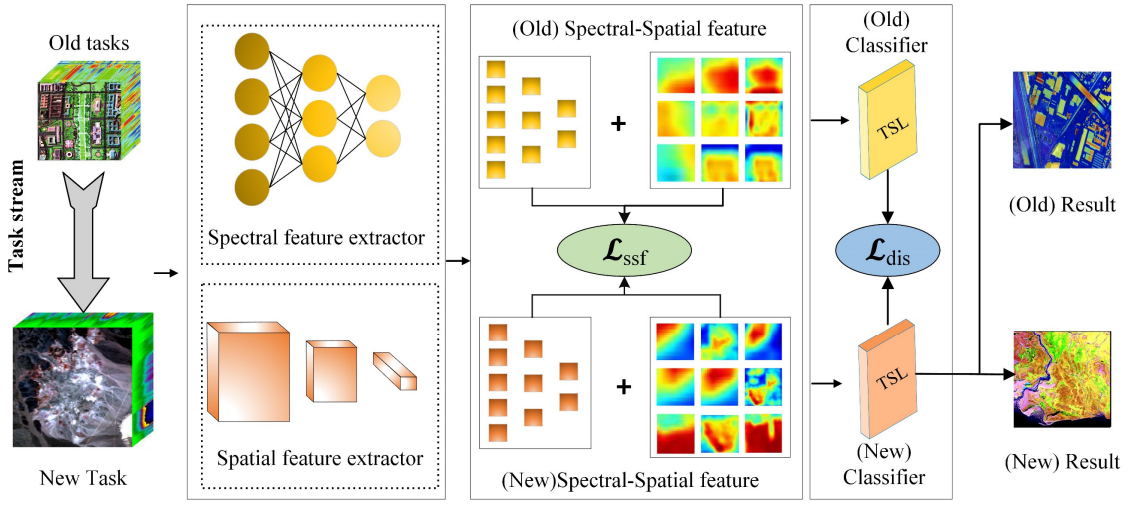

Fig. 1. Framework diagram of our proposed model. The loss function of the network consists of task cross-entropy loss, spatial-spectral loss, and targets loss. Old task datasets are memorized input to the new task model with new task datasets. The cross-entropy loss is used to learn new task datasets; spatial-spectral loss is used to preserve learned spatial and spectral knowledge, and target loss is applied to preserve higher level semantic information of the old model.

network, triplet network, and n-pair network. For instance, Zhao et al. [38] proposed a Siamese neural network-based method for HSI classification, which employs two identical pixels as positive and negative sample input for model training. Similarly, Huang and Chen [39] presented a dual-path siamese CNN (Dual-SCNN), which generates positive and negative samples with data augmentation for contrastive loss training. These approaches successfully distinguished different objects and obtained meaningful information by the share-weights Siamese network. Inspired by the Siamese network, the triplet network approach formed positive, negative, and anchor samples, which focused on measuring the similarity or dissimilarity between anchor samples and positive samples and between anchor samples and negative samples, respectively. Deng et al. [40] adopted a triplet network based on deep metric learning for HSI classification, which trained two similar samples and one different sample in each triplet simultaneously. An online triplet loss-based deep metric learning method proposed by Dong et al. [41], effectively improved the classification performance of HSIthrough the same training strategy. In addition, the utilization of quadruplet loss and even n-pair loss has greatly improved the performance of the model. Considering that triplet loss only utilized positive and negative samples limits the generalization ability of the model. Zhang et al. [42] introduced another sample based on the triplet loss to create quadruplet loss for model training. These methods based on quadruplet losses, however, create a dramatic increase in training samples, and the multiplication of training samples increases computational time causes and requires too much storage space during the training process.

III. Proposed Method

To enable recurrent classification for a variety of hyperspectral datasets, we propose a continual learning scheme to distillate the spectral-spatial, class-specific feature, without "forgetting." The flowchart of the proposed method is shown in Fig. 1. First, we propose to use the deep-learning-based spectral-spatial feature extraction framework. Then, as shown in Fig. 2, we add the spectral-spatial feature distillation, which is calculated by applying the metric learning scheme. In addition, the KD is also measured with the similarity term generated from metric learning. Finally, the parameters of the classification matrix are updated by considering both the spectral-spatial feature similarity and the classification term.

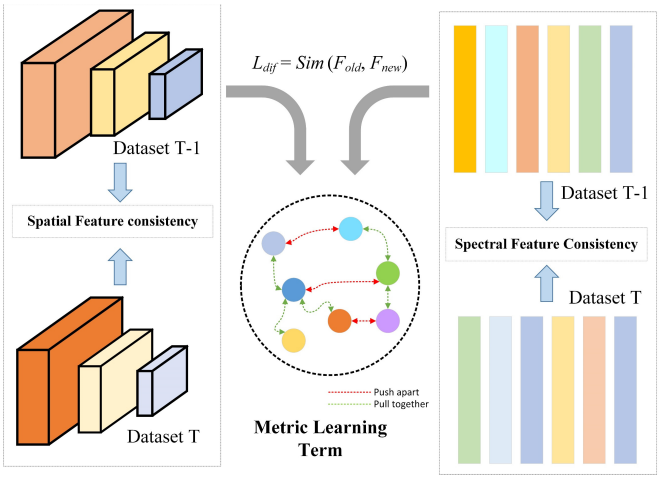

Fig. 2. Metric learning for spatial and spectral knowledge retention.

A. Metric Learning With Spectral-Spatial Feature Distillation

In order to achieve the purpose of feature distillation, CNNs are applied to extract spectral-spatial features. The HSI $X \in \mathbb{R}^{W\times H\times B}$ can be represented by a 3-D data cube having a width of W, a height of H, and a number of spectral bands, B. In general, HSI classification tasks primarily rely on precise spectral information to achieve superior performance. Assuming that the spectral vector $x_{i,j} \in \mathbb{R}^{1\times 1\times B}$, where i, j represent the column and row of the original image and B

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denotes the number of channels of HSI, these parameters are used for the convolution in the spectral dimension to obtain the spectral features. To further obtain the distinguishing features of the input samples, the cropped 3-D tensor $P_1 \in \mathbb{R}^{w\times h\times b}$, defines a spatial patch composed of a central pixel and neighbors, where w, h, and b denote the width, height, and the number of channels in a patch, respectively, these parameters are used to capture spatial context information. The training set contains a collection of 3-D tensors $P = \{P_1, P_2, P_3, \ldots, P_N\}$. The spatial patch and spectral vector corresponding labels are defined as $Y = c, c \in (1, 2, \ldots, C)$, where C denotes classes of ground truth. The label of each patch is represented by the central pixel, and the label of the spectral vector is the category of the corresponding pixel. Hence, in our implementation, we designed a model, which consists of spatial-feature extractor coupled with a spectral-feature extractor. The 1-D spectral vector and 3-D spatial patches can be used as inputs to the deep CNN to extract the spectral and spatial features and concatenate them as inputs to the final classification layer. In this scene, the robust features so obtained will further improve the performance of the model for HSI classification.

To capture the spectral-domain information of land cover from a large number of hyperspectral bands, a deep spectral dimension CNN module is designed. Typically, CNN mainly consists of a convolution layer, a pooling layer, and a nonlinear layer. In detail, the module is fed with a spectral pixel vector $x_{i,j}$ to obtain hierarchical features. The calculation of lth feature map of the spectral vector can be expressed as

$$f_{\text{spe}}^{l} = \sigma\left(\sum_{i=1}^{k} W_i^l * f_{\text{spe}}^{l-1}\left(x_{i,j}\right) + W_b^l\right) \tag{1}$$

where $*$ denotes the convolution operation, f_{spe}^{l-1} represents the output feature map of the $l-1$th layer, W_i^l and W_b^l are the weight and bias parameters that are to be trained, $\sigma(\cdot)$ represents the non-linear activate functions that are rectified to a linear unit (ReLU).

In the CNN module, the pooling layer is used to decrease the redundant hyperspectral bands, and it can significantly reduce the complexity and minimize the number of training parameters. The feature map after the pooling operation can be expressed as

$$f_{\text{spe}}^{l} = \text{Pool}\left(\sigma\left(\sum_{i=1}^{k} W_i^{l*} f_{\text{spe}}^{l-1}\left(x_{i,j}\right) + W_b^l\right)\right) \tag{2}$$

where Pool$(\cdot)$ represent max pooling. The obtained spectral domain features become more abstract and discriminate with the lth convolution operation.

Simultaneously, in order to extract internal spatial context information of HSI, another 2-D CNN module is designed. The spatial-feature extraction module and the spectral-feature extraction module have similar structures; the module is composed of a convolutional layer, an activation layer, and a pooling layer. The module takes the 3-D tensor $P_n \in \mathbb{R}^{w\times h\times b}$ as input and uses the kernel to perform window sliding in the spatial dimension over the patch. The corresponding convolution operation can be expressed as

$$f_{\text{spa}}^{l} = \sigma\left(\sum_{i=1}^{k} W_i^l * f_{\text{spa}}^{l-1}(P_n) + W_b^l\right) \tag{3}$$

where $*$ denotes the convolution operation, f_{spa}^{l} is the feature map from the current lth layer, f_{spa}^{l-1} represents the output feature map of the previous layer, W_i^l and W_b^l are the trainable weight and bias parameters, and $\sigma(\cdot)$ represents the ReLU activation function.

The final feature output is obtained through a maximum pooling operation

$$f_{\text{spa}}^{l} = \text{Pool}\left(\sigma\left(\sum_{i=1}^{k} W_i^l * f_{\text{spa}}^{l-1}(P_n) + W_b^l\right)\right). \tag{4}$$

After these processes, the spectral and spatially hierarchical deep features of the HSI were the results obtained. These two features can be fused to generate more robust features. More information can be preserved, which further improves the classification performance. To simplify the calculation, the concatenation of spectral and spatial representative features, extracted by the 1-D module and 2-D module, are used as inputs to the softmax layer for classification. The process of feature fusion is represented as

$$f_{\text{out}} = \text{cat}\left(f_{\text{spa}}^{l}, f_{\text{spa}}^{l}\right) \tag{5}$$

where f_{out} denotes the features that will be input to the classifier layer, and cat$(\cdot)$ represents the feature concatenation operation.

In the continuous learning process, the model, however, inevitably "forgets" significant feature information of the historical data when new data are added. The most important purpose of continual learning is to maximize the retention of previously learned historical knowledge while learning new classifications employing new data. Feature metric learning becomes critical to preserving historical knowledge. Feature metric learning helps to minimize the distance between features extracted from the old and new models in the feature space and maintains the consistency of the performance of the old and new models. The most common measure of feature similarity is cosine similarity, which is calculated from the cosine angle multiplication of two vectors. Therefore, the cosine similarity of the deep features between the old and new models can be expressed as

$$\text{sim}(f_{\text{old}}, f_{\text{new}}) = \frac{\Sigma_{i=1}^{n} f_{\text{old}} \times f_{\text{new}}}{\sqrt{\Sigma_{i=1}^{n}(f_{\text{old}})^2} \times \sqrt{\Sigma_{i=1}^{n}(f_{\text{new}})^2}} \tag{6}$$

where f_{old} and f_{new} denote the deep features extracted by the historical task model and the new task model, respectively, and n is the length of the feature.

Since the feature extraction model designed in this article contains both the spectral feature extractors and the spatial feature extractors, it is necessary to measure the consistency between the old and new models in terms of the above two features separately. The two feature metric is calculated

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as follows:

$$\text{spatial_sim} = \frac{1}{NL}\sum_{n=1}^{N}\sum_{L=l}^{L}\text{sim}\left(F_{\text{old}}^{l}(P_n),\quad F_{\text{new}}^{l}(P_n)\right) \tag{7}$$

$$\text{spectral_sim} = \frac{1}{NL}\sum_{n=1}^{N}\sum_{L=l}^{L}\text{sim}\left(F_{\text{old}}^{l}(x_{i,j}),\quad F_{\text{new}}^{l}(x_{i,j})\right) \tag{8}$$

$$\text{sim_metric} = \lambda\cdot\text{spatial_sim} + (1-\lambda)\cdot\text{spectral_sim} \tag{9}$$

where N is the number of training samples in a batch, L is the number of intermediate model feature layers, λ denotes similarity weights that are normally set to 0.5, $F_{\text{old}}^{l}(\cdot)$ and $F_{\text{new}}^{l}(\cdot)$ denote the lth layer features extracted by the historical model and the new model, respectively.

Consequently, in order to maintain the learned knowledge in the new hyperspectral classification model, the metric loss function is constructed by maximizing the consistency of the spectral and spatial features of the old and new models, respectively. The feature consistency loss function is designed as follows:

$$L_{\text{met}} = \min\left(1-\text{sim_metric}\right). \tag{10}$$

The value of the loss function ranges from 1 to 0 because the value of the cosine similarity ranges from 0 to 1. The closer the value of the loss function is to 0, the more consistency exists between the old and new model features.

B. Metric Learning With Class Distillation

Similarity matching of different hierarchical feature maps between the historical model and the new model preserves historical knowledge. In addition, the prediction results should be equally consistent between the old and new models. Accordingly, targets divergence loss for latent knowledge is designed as follows:

$$L_{\text{target}} = \frac{1}{N}\sum_{n=1}^{N}\left(F_{\text{new}}\left(x_{i,j},P_n,\theta_{\text{new}}\right) - F_{\text{old}}\left(x_{i,j},P_n,\theta_{\text{old}}\right)\right)^2 \tag{11}$$

where N is the number of training samples in a batch, F_{new} and F_{old} represent the outputs of the old and new model, respectively, and θ_{new} and θ_{old} are the parameters of these models. Compared with feature-based metric loss, the target loss effectively obtains knowledge by minimizing the difference between predictions of the old and new models.

In HSI classification tasks, the cross-entropy loss function is often used to measure the difference in the probability distribution between the model prediction and the ground truth. For the input spectral pixel vector $x_{i,j}$ and spatial patch P_n that correspond to the true label y_c, the softmax layer transforms the last acquired features f_{out} into the corresponding categories, and the process is represented as follows:

$$p(\hat{y}=c|f_{\text{out}}) = \frac{\exp\left(W^s\ \cdot\ f_{\text{out}}+W_b^s\right)}{\sum_{c=1}^{C}\exp\left(W^s\ \cdot\ f_{\text{out}}+W_b^s\right)} \tag{12}$$

where $\hat{y}$ is the prediction result of model, C represents the number of land cover classes, and W^s and W_b^s are weights and bias parameters in the softmax layer.

Therefore, the current task cross-entropy loss function formula is shown as follows:

$$L_s = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C}y_c\log(p(\hat{y}=c|f_{\text{out}})) \tag{13}$$

where N denotes the number of training samples in a batch. The category c then keeps growing with the new task updates. For example, the model has been trained on all previous $t-1$ hyperspectral tasks with corresponding categories $C_1, C_2, \ldots, C_{t-1}$. Then for the current task t, the final category C is defined as $C=\sum_{t=1}^{T}C_t$, the sum of all task categories.

In a sequence of constantly changing tasks, the continual learning loss function used for the optimization of the current hyperspectral task model can be formulated as follows:

$$L_{\text{total}} = L_{\text{met}} + L_{\text{target}} + L_{\text{entro}}. \tag{14}$$

It can be seen that designed continual learning loss for hyperspectral classification tasks consists of the feature consistency term, the targets term, and the cross-entropy for current. Among them, the feature consistency term L_{met} is used for spectral features and spatial feature hierarchical knowledge retention. This term preserves knowledge from the previous models by maintaining consistency between old and new models' extracted spectra and spatial features. For target divergence loss term, higher level semantic information transferred easily and efficiently from the historical model to the current model. As a typical loss function, the cross-entropy term is used to learn new knowledge, which measures the distance between the prediction and the ground truth of the model continually during the learning process. The hyperspectral continual learning procedure is described in Algorithm 1.

IV. Experiments

In this section, two hyperspectral datasets utilized in this article are described. Also, the implementation details in the experiments are introduced. In order to evaluate the classification accuracy with other benchmark methods, overall accuracy (OA), average accuracy (AA), and kappa coefficient (K) are utilized. We further evaluate the effectiveness of our model on continual learning tasks, the AA Ave and average forgetting accuracy (AF) are introduced.

A. Data Description

1) Pavia University Data Set: The Pavia university dataset was captured by Reflective Optics System Imaging Spectrometer (ROSIS-03) optical sensor in the area near the university of Pavia, Italy. The ROSIS-03 sensor provides multiband images containing 115 spectral bands in the wavelength range of 0.43–0.86 μm. In practice, twelve bands influenced by noise were removed, leaving 103 spectral channels. The image size is 610 $\times$ 340 pixels with a spatial resolution of 1.3 m. The dataset contains nine categories of urban land cover features, including meadows, bare soil, gravel, and other urban features,

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Algorithm 1 The proposed life-long learning framework for hyperspectral classification

1. **Require**:
   Updating data sets D_t for task T, $t \in 1, 2, \ldots, T$
   the network $F(x_{i,j}; P_n; \theta_t)$, $t = 1, 2, \ldots, T$
   random initialized parameters W^* and W^b
   spectral_sim(·) and spatial_sim(·) are cosine similarity

2. **for** t = 1, 2, …, T do
3. $D_{t+1} \leftarrow D_t$
4. Generate pixel vector $x_{i,j}$ and patch P_n
5. **for** s=1, 2, …, S do
6. **for** m = 1, 2, …, M do
7. sample a mini-batch M from D_t
8. **for** all $x_{i,j}$ and $P_n \in M$ do
9. $spectral_sim(x_{i,j}, P_n)$
10. $spatial_sim(x_{i,j}, P_n)$
11. metric loss L_s
12. target loss L_l
13. cross-entropy loss L_C
14. **end for**
15. update parameters W_t^* and W_t^b via (14)
16. **end for**
17. updating $W_{t+1}^* \leftarrow W_t^*$
18. updating $W_{t+1}^b \leftarrow W_t^b$
19. **end for** #
20. **end for**
21. Return Model for current task T $F(x_{i,j}; P_n; \theta_t)$

with only 42 776 pixels labeled in total. Table I shows the number of annotated pixels corresponding to these nine categories, while the remaining 2 164 624 pixels are background features without any reference information.

2) Indian Pines Data Set: The second data set was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor with a total of 220 spectral channels in the wavelength range from 0.4 to 2.5 μm. This HSI was captured at the Indian Pines test site in Northwestern Indiana, USA. In general, among these 220 spectral channels, only 200 bands are normally used because the other 20 spectral bands are absorbed by water rather than being reflected. In this article, the entire 220 channels are employed for the experiments. The image has a size of 145 × 145 pixels and a spatial resolution of 20 m. There are only 10 249 pixels in total with annotations, while the remaining 10 776 pixels are background pixels. The entire image area is covered with 16 different categories of ground cover, and most of them are crops, such as soybeans, wheat, and corn. Table I shows the number of different feature types and their corresponding annotated pixels. It can be seen that the number of annotated samples in each category is imbalanced, which poses a challenge for accurate classification.

3) Salinas Scene Data Set: The third data set is the Salinas scene, which was acquired by the AVIRIS sensor over the Salinas Valley, California. The image size is 512 × 217 pixels with a spatial resolution of 3.7 m per pixel. This data set contains 224 spectral bands with 10 nm spectral resolution. Similar to the Indian pines data set, in order to avoid the loss of spectral information, we also retained the 20 water absorption bands in our experiments. The dataset contains 16 labeled categories of agricultural land covers, such as weeds, fallow, corn, and vineyards. The different land covers and their corresponding number of annotated pixels are shown in Table I.

4) Houston Data Set: The last dataset is located in the University of Houston campus and the neighboring urban area, which was released in the IEEE Geoscience and Remote Sensing Society (GRSS) data fusion contest in 2013. The image size is 349 × 1905 pixels and 144 spectral bands, the spatial resolution is 2.5 m per pixel and it contains 15 classes in all, such as tree, soil, residential, road, etc. In this dataset, 15 029 annotated pixels and 649 816 pixels are backgrounds without any labeled information.

B. Evaluation and Experimental Settings

For the above four data sets, the spectral dimensions are 103, 220, and 224, respectively. To generate universal spectral inputs, we performed spectral alignment operations on these four data sets. Specifically, to avoid information loss, the number of spectral bands for all datasets are replicated to have the same spectral dimension for the 1D-CNN model. Simultaneously, we cropped images to 18 × 18 pixel squares as the input to the 2D-CNN model in order to obtain the spatial features of the HSI.

For the four data sets, to ensure the effectiveness of the model, the number of samples was divided as follows, we split the four data sets with 1% labeled samples for training and another 20% labeled samples for validation. Since the Indian pines data set has a small number of samples, we apply a larger number of samples for partial categories. During the continual learning session, we set the training data of the historical data set to be retained and engaged in the training of the new task together with the new dataset. We train the model with Adaptive Moment Estimation (Adam) with a weight decay of 0.001 and a learning rate of 0.001. We use 100 epochs to train the model. Batch normalization and a dropout mechanism are adopted. In the experiment, a server with OS centos of version 7, 256G RAM, and two Tesla V100 GPUs with 16 GB memory are used for experiments. To quantitatively estimate the classification ability of the proposed model with other benchmark methods, OA, AA, and K are computed.

To further explain how the model transfers knowledge, in addition to the final validation accuracy of the model, we introduce the AA and average AF to measure the model's learned and forgotten knowledge [43]. Formally, we assume that $acc_{i,j}$ denotes the validation accuracy of the model on task j after training on task i. The number of tasks is defined as T in total, the AA Ave can be represented as the average of the accuracies on task j over the continuous learning process from task 1 to task i. A higher AA means that the model performed better using continuous learning. The formula is expressed

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TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES OF FOUR DATASETS

Pavia				Indian pines				Salinas				Houston			
Class	Total	Train	Test	Class	Total	Train	Test	Class	Total	Train	Test	Class	Total	Train	Test
0	6631	66	1326	0	1428	1	7	0	2009	20	401	0	1251	14	276
1	18649	186	3729	1	830	16	315	1	3726	37	745	1	1254	11	234
2	2099	20	419	2	237	8	128	2	1976	19	395	2	697	9	154
3	3064	30	612	3	483	1	37	3	1394	13	278	3	1244	10	259
4	1345	13	269	4	730	7	83	4	2678	26	535	4	1242	13	239
5	5029	50	1005	5	478	8	167	5	3959	39	791	5	325	5	59
6	1330	13	266	6	972	3	7	6	3579	35	715	6	1268	16	259
7	3682	36	736	7	2455	2	94	7	11271	112	2254	7	1244	10	277
8	947	9	189	8	593	1	5	8	6203	62	1240	8	1252	12	250
-	-	-	-	9	205	8	196	9	3278	32	655	9	1227	14	211
-	-	-	-	10	1265	21	530	10	1068	10	213	10	1235	5	251
-	-	-	-	11	386	2	133	11	1927	19	385	11	1233	15	233
-	-	-	-	12	93	22	42	12	916	9	183	12	469	4	94
-	-	-	-	13	46	2	253	13	1070	10	214	13	428	3	75
-	-	-	-	14	28	3	35	14	7268	72	1453	14	660	9	134
-	-	-	-	15	20	1	17	15	1807	18	361	-	-	-	-

TABLE II
OA OF DIFFERENT WEIGHTS OF SPATIAL AND SPECTRAL

Spa:Spe	1:9	2:8	3:7	4:6	5:5	6:4	7:3	8:2	9:1
Step1	93.13	93.13	93.13	93.13	93.13	93.13	93.13	93.13	93.13
Step2	84.11	83.46	83.64	84.11	85.08	84.74	85.36	83.06	84.01
Step3	90.02	88.83	81.37	87.85	88.94	86.35	87.11	81.73	86.54
Step4	85.77	82.84	87.77	75.94	78.51	86.37	87.80	88.95	83.04
Ave	88.26	87.06	86.48	85.26	86.42	87.65	**88.35**	86.72	86.68

TABLE III
OA OF DIFFERENT WEIGHTS OF FEATURE KNOWLEDGE AND TARGET KNOWLEDGE

Fea:Target	1:9	2:8	3:7	4:6	5:5	6:4	7:3	8:2	9:1
Step1	93.13	93.13	93.13	93.13	93.13	93.13	93.13	93.13	93.13
Step2	75.12	83.76	86.67	87.93	85.03	83.40	83.65	83.39	83.71
Step3	90.83	84.95	80.18	88.82	86.61	86.56	88.01	88.16	89.13
Step4	89.08	83.15	89.19	88.12	86.35	83.15	83.36	86.08	88.22
Ave	87.04	86.25	87.29	**89.50**	87.78	86.56	87.04	87.69	87.79

as follows:

$$\mathrm{Ave}(i) = \frac{1}{i}\sum_{j=1}^{i} \mathrm{acc}_{i,j}. \tag{15}$$

The average AF indicates the forgotten elements of task j by the model after training on task i. Average forgetting for task i is calculated as follows:

$$\mathrm{AF}(i) = \frac{1}{i-1}\sum_{j=1}^{i-1} f_{i,j}$$
$$\text{where} \quad f_{k,j} = \max_{l\in\{1,\ldots,k-1\}} \left(\mathrm{acc}_{l,j}\right) - \mathrm{acc}_{k,j} \quad \forall j < K \tag{16}$$

and where $\max_{l\in\{1,\ldots,k-1\}}(\mathrm{acc}_{l,j})$ denotes the best validation performance of the model on task j with training on task k, and $\mathrm{acc}_{k,j}$ denotes the validation accuracy on task j after training on task k. A lower $\mathrm{AF}(i)$ indicates that the model has remembered more knowledge about the previous task in the sequence of continual learning tasks.

C. Ablation Experiments

To evaluate our proposed continuous learning approach accurately, we have completed several ablation experiments on the loss function designed in this article for continual learning on HSI classification. In order to evaluate the effectiveness of the different modules, we set four tasks on our data sets Pavia, Indian pines, Salinas, and Houston. The results of these ablation experiments are shown in the following tables and figures.

First, to explore the effectiveness of spatial and spectral features distillation strategies for knowledge retention, we set the spatial loss and spectral loss with different weights from 0.1 to 0.9. For fair comparison, the first-step model parameters are frozen for conducting experiments. As shown in Table II, the spatial features provide informative knowledge with the weight growth of spatial distillation module, the optimal result is reached when the spatial and spectral features module is 7:3; however, the model suffers from instability problem with the decreasing weight of the spectral features. This result indicates that spatial-spectral features enable the model performs robust during continual learning. In this article, we set the weight ratio of the spatial and spectral loss to 7:3.

We further analyzed the effect of merged feature knowledge and target knowledge during the continual learning process of the model. As shown in Table III, the model performed well when the feature distillation and targets distillation contribution similarly during the four-step continuous learning tasks. With larger weights of target knowledge modules, the model fails to obtain stabilized results at different steps of the task in continuous learning, and as the weights of the feature knowledge retention module become larger, the AA of the four-step continual tasks gradually declines. To balance the effectiveness of feature KD and target knowledge preservation, we set the weight ratio as 4:6 in this article.

The results in Table IV clearly demonstrate how the combination of cross-entropy, spatial-spectral loss, and class loss enables the model to perform best. The fusion of spatial textural features and spectral information enables the model to obtain more robust features. Compared with using a single spatial or spectral loss module, better results are produced in the later tasks with the use of both spatial and spectral

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TABLE IV

EVALUATION OF DIFFERENT COMBINATION OF LOSS MODULE FOR CONTINUAL LEARNING TASKS ON PAVIA, INDIAN PINE, SALINAS, AND HOUSTON DATASETS

#	Entro	Spa	Spe	Class	L1	L2	Step1			Step2			Step3			Step4		
							OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
1	√	√					96.06	93.12	0.95	93.06	93.34	0.93	91.33	74.39	0.91	86.24	72.29	0.86
2	√		√				96.06	93.12	0.95	94.45	89.89	0.93	91.19	74.64	0.91	84.79	**75.72**	0.84
3	√	√	√				96.06	93.12	0.95	95.19	93.62	0.95	91.62	75.51	0.91	87.24	72.29	0.87
4	√	√	√	√			96.06	93.12	0.95	**95.29**	**93.79**	**0.95**	**91.89**	**75.56**	**0.91**	**88.74**	74.65	**0.88**
5	√	√	√	√	√		96.06	93.12	0.95	88.82	86.74	0.87	88.96	75.35	0.88	87.87	73.34	0.87
6	√	√	√	√		√	96.06	93.12	0.95	95.00	92.49	0.94	88.42	72.80	0.88	88.32	74.43	0.88

TABLE V

ABLATION EXPERIMENTS FOR DIFFERENT LAYER FEATURE DISTILLATION ON FOUR STEPS CONTINUAL LEARNING TASKS ON PAVIA, INDIAN PINE, SALINAS, AND HOUSTON DATASETS. THE FOUR DATASETS USED 1% ANNOTATED SAMPLES FOR TRAINING AND 20% SAMPLES FOR TESTING

		Step1	Step2		Step3			Step4			
		Pavia	Pavia	Indian pines	Pavia	Indian pines	Salinas	Pavia	Indian pines	Salinas	Houston
Layer1	OA	96.06	95.98	45.92	92.85	69.20	92.26	93.55	64.47	93.64	80.10
	AA	93.12	95.16	22.34	88.34	56.83	94.87	90.02	51.43	95.15	79.02
	K	0.95	0.95	0.35	0.91	0.64	0.91	0.92	0.59	0.93	0.78
Layer2	OA	96.06	97.93	57.15	93.20	37.29	82.41	94.94	68.13	95.06	75.34
	AA	93.12	97.60	52.10	88.47	33.72	86.35	90.76	51.26	96.43	75.18
	K	0.95	0.97	0.50	0.91	0.28	0.80	0.93	0.63	0.94	0.73
Layer3	OA	96.06	98.92	59.61	90.01	60.24	70.75	94.18	63.79	94.07	78.50
	AA	93.12	98.13	59.70	85.98	59.16	70.38	90.44	47.64	95.17	77.11
	K	0.95	0.99	0.58	0.87	0.60	0.67	0.92	0.58	0.93	0.77
Layer4	OA	96.06	95.13	69.79	94.39	69.40	93.64	92.45	67.30	93.69	77.87
	AA	93.12	95.52	68.99	92.27	57.91	94.30	87.25	55.55	95.27	76.83
	K	0.95	0.94	0.68	0.93	0.64	0.93	0.90	0.62	0.93	0.76
All layers	OA	96.06	93.55	64.47	92.04	65.35	90.72	93.19	61.84	91.59	79.13
	AA	93.12	88.87	46.49	85.61	48.34	93.43	87.68	47.03	94.56	78.11
	K	0.95	0.92	0.58	0.90	0.59	0.90	0.91	0.55	0.91	0.77

features distillation. The mutual constraint of spatial and spectral features by spatial-spectral loss allows the model to retain more historical knowledge than either one alone. They also cause the output of the final layer of the model to contain higher-level semantic information of the hyperspectral data. Thus, the introduction of target loss over spatial-spectral loss ensures efficient knowledge transfer to a new task. The fourth row of the ablation results in Table IV, OA, AA, and Kappa demonstrate the optimal performance of this approach. In addition, we evaluated the effectiveness of the regularization term for continual learning by adding L1 and L2 terms. The fifth and sixth rows show the effects of introducing L1 and L2 regularization loss, respectively. It can be seen by the further improvement of the model is limited in step 2, step 3, and step 4 tasks. To sum up, the above results prove that our proposed method effectively alleviates the problem of catastrophic forgetting while constantly updating the model.

In addition, as shown in Fig. 3, the AA curve and the average AF curve of different loss modules demonstrate the effect of the different loss modules on the four steps. For these accuracy results plots, the red line shows our proposed method achieves slightly higher performance, which utilizes entropy loss, spatial-spectral loss, and targets loss. Compared with the model without target loss, our option offers better results when the history module updates to a new module. This is because the predictions output from the final layer of the model incorporates high-level and abstract information, which effectively enables the next step of the model to learn new knowledge from the new dataset. Although the spatial and

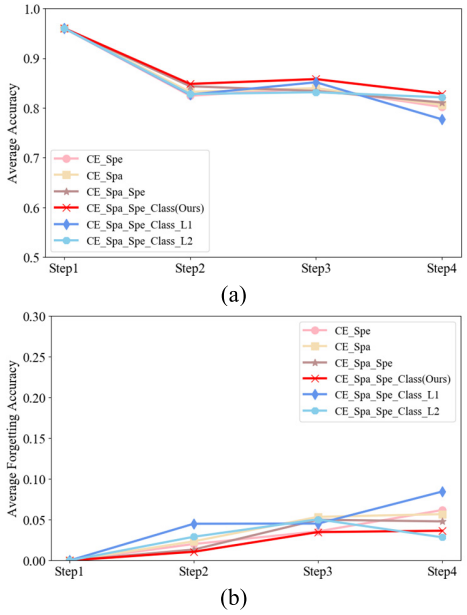

Fig. 3. AA curve (a) and average AF curve (b) for four steps of continual learning tasks.

spectral feature losses generate robust features and enable the potential knowledge transfer to the new model, it is limited in learning new knowledge quickly. Simultaneously, the parameter penalty term L1 and L2 regularization demonstrate more about protection critical weights while ignoring the acquisition

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TABLE VI
CLASSIFICATION RESULTS FOR EACH CATEGORY OF INDIVIDUAL DATASET OF THE FOUR-STEP CONTINUAL LEARNING TASKS

	Step1	Step2		Step3			Step4			
Class	Pavia	Pavia	Indian pines	Pavia	Indian pines	Salinas	Pavia	Indian pines	Salinas	Houston
0	94.62	99.39	100.00	90.75	14.29	100.00	92.80	85.71	100.00	94.47
1	99.48	99.94	76.83	99.54	83.49	98.31	47.98	74.60	54.05	90.12
2	85.82	96.71	95.31	86.84	25.00	95.53	86.84	17.97	85.11	63.01
3	97.78	98.73	78.38	98.10	13.51	97.21	98.89	16.22	98.95	82.38
4	98.88	98.51	100.00	97.77	89.16	95.62	95.91	20.48	96.13	97.84
5	98.82	97.56	94.01	99.64	92.22	100.00	99.37	78.44	98.23	26.15
6	74.13	92.66	100.00	76.92	0.00	99.16	65.73	0.00	98.05	63.78
7	89.81	88.37	94.68	93.41	6.38	83.98	86.09	12.77	72.87	46.91
8	98.71	100.00	100.00	98.28	20.00	98.61	100.00	80.00	98.45	49.22
9	--	--	96.43	--	39.29	98.40	--	77.55	93.12	62.10
10	--	--	64.72	--	61.89	95.45	--	55.85	92.73	81.89
11	--	--	87.22	--	10.53	97.25	--	11.28	53.30	63.78
12	--	--	100.00	--	35.71	99.41	--	54.76	99.41	67.39
13	--	--	99.60	--	89.33	98.25	--	85.77	97.37	67.90
14	--	--	85.71	--	40.00	94.66	--	40.00	86.47	67.69
15	--	--	100.00	--	88.24	95.71	--	100.00	57.94	--
OA	96.06	97.90	84.43	96.06	59.79	94.67	74.31	57.00	84.68	71.01
AA	93.12	96.88	92.06	93.47	44.31	96.72	85.96	50.71	86.39	68.31
Kappa	0.95	0.97	0.82	0.95	0.53	0.94	0.69	0.51	0.83	0.69

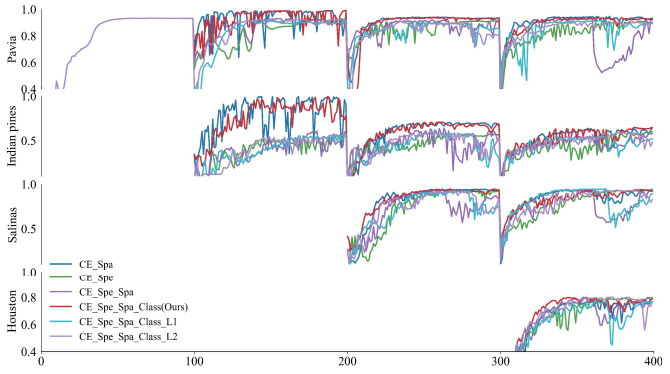

Fig. 4. Testing accuracy curve of model for different combinations of loss functions. We evaluate the performance of the model for the four combinations of loss functions on Pavia, Indian pine, and Salinas datasets. The four datasets used 1%, annotated samples for training and 20% samples for testing.

of new knowledge. Similarly, the average forgetting curve proves that the knowledge learned from the historical model is effectively maintained, especially for the L2 parameter regular term.

In order to verify the accuracy performance of the models with different combinations of loss functions during the training process, we plot the overall validation accuracy curves of different combinations of loss functions for each training process. The performance curves with different modules are shown in Fig. 4. The model exhibits significant fluctuations in the second stage of the task when using only cross-entropy and spatial loss. Integrated spatial and spectral features provide more robust features. The method performance recovery slowly, which employ cross-entropy and spatial-spectral loss during task switching. It shows that by only utilizing entropy loss and spatial-spectral loss are unable to learning knowledge quickly from novel data. The combination of the underlying cross-entropy loss, spatial-spectral loss, and target loss show high and robust accuracy performance in the four-step process. When the L1 or L2 regularization terms are introduced, slightly decreased performances are shown in the second, third, and fourth step processes. This demonstrates that the L1 and L2 norms limit the model from further improvements. And the model performances are reduced in the final phases of the training process. Consequently, the task cross-entropy loss module, spatial-spectral loss module, and targets loss module are used for model training in this article.

To investigate the effect of different layer features of the feature distillation module over the continual learning process, we additionally designed ablation experiments using different layer feature distillation. As shown in Table V, Layer 1–4 represents the different layers' features for features distillation, and all layers indicate that features of all layers are employed for feature KD. Compared with using all layer features for spatial-spectral feature distillation, the highest OA is obtained by distillation utilizing only the fourth layer of features at the first step and second step tasks. It is because shallow spatial-spectral features incorporate a lot of redundant information, which makes the model suffers more uncertainty during continual learning. Although it does not reach the optimal performance in the fourth step, it can balance multiple steps using only the fourth layer of feature distillation to achieve satisfactory accuracy performance in the overall continual learning process. In summary, the informative spatial-spectral knowledge could be effectively transferred to future tasks by utilizing the deep abstract layer features distillation.

D. Comparison With Baseline Methods

In order to demonstrate the classification capability of our model, comparison experiments were carried out to compare with other more classical continuous learning models. Learning without Forgetting (LwF) [44], Synaptic Intelligence (SI) [45], Context-dependent-Gating (XdG) [46], EWC [15], and Online EWC [47] and fine-tuning were used to perform comparison experiments. In this article, the parameter λ for LWF and EWC is set to 1 and 50, respectively, and the λ

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TABLE VII

COMPARISON RESULTS OF INDIVIDUAL DATASET OF DIFFERENT METHODS ON FOUR STEPS CONTINUAL LEARNING TASKS ON PAVIA, INDIAN PINE, SALINAS, AND HOUSTON DATASETS. THE FOUR DATASETS USING 1% ANNOTATED SAMPLES FOR TRAINING AND 20% SAMPLES FOR TESTING

Method	Dataset		EWC*	EWC-online*	LWF*	SI*	XDG*	Fine-tuning	Ours2
Step 1	Pavia	OA	92.89±0.29	92.93±0.34	92.94±0.36	92.87±0.34	92.39±0.7	92.89±1.68	94.48±1.15
		AA	88.95±0.32	89.04±0.25	89.01±0.35	88.93±0.3	88.09±0.39	90.34±2.46	91.51±1.85
		K	0.91	0.91	0.91	0.91	0.9	0.91	0.93
Step 2	Pavia	OA	91.6±0.53	91.49±0.58	92.01±0.53	92.06±0.55	90.96±1.09	91.89±15.29	94.99±4.49
		AA	86.99±0.95	86.56±1.03	87.99±0.59	87.59±0.77	84.82±1.76	94.66±9.01	96.47±1.85
		K	0.89	0.89	0.9	0.9	0.88	90.58±17.66	93.72±5.55
	Indian pines	OA	54.31±3.51	55.31±3.13	50.45±5.33	56.04±4.71	45.45±7.12	70.13±3.47	71.69±1.54
		AA	37.03±3.0	37.76±3.34	33.3±3.64	39.31±4.64	28.78±5.48	70.82±2.65	71.92±2.71
		K	0.47	0.48	0.42	0.49	0.37	0.70	0.70
Step 3	Pavia	OA	86.63±0.9	88.08±0.49	89.78±1.63	88.32±2.33	87.49±2.24	93.69±3.89	93.20±2.78
		AA	79.96±3.18	82.12±1.89	84.5±1.35	83.29±2.64	79.59±3.02	92.45±2.76	88.25±2.25
		K	0.83	0.85	0.87	0.85	0.84	91.97±4.83	0.91
	Indian pines	OA	46.76±5.25	49.2±3.58	48.88±3.28	51.42±3.62	43.29±6.5	62.55±2.12	62.67±4.15
		AA	32.1±2.67	33.68±3.97	32.74±2.82	33.78±3.49	27.43±4.9	46.74±2.61	44.99±3.01
		K	0.39	0.41	0.4	0.43	0.34	0.55	56.59±4.73
	Salinas	OA	87.53±3.32	89.52±1.26	84.19±1.8	90.83±1.33	85.97±2.48	90.14±2.77	90.56±4.4
		AA	88.74±3.06	91.52±1.88	83.38±4.25	92.21±2.68	83.91±4.0	92.05±1.57	92.28±4.08
		K	0.86	0.88	0.82	0.9	0.84	0.89	0.90
Step 4	Pavia	OA	84.46±3.28	86.59±1.02	89.01±0.47	87.7±0.89	87.22±1.06	90.89±5.76	88.16±7.88
		AA	76.27±4.54	79.73±2.34	82.89±0.84	82.19±0.68	81.16±2.43	88.19±6.11	87.48±4.03
		K	0.8	0.83	0.86	0.84	0.84	0.84	0.86
	Indian pines	OA	46.36±5.53	43.41±4.05	47.76±3.27	51.66±5.04	48.9±5.6	59.8±5.4	61.75±7.45
		AA	31.12±3.68	29.83±2.84	32.08±1.98	36.34±3.73	30.6±3.9	45.85±3.19	48.12±5.4
		K	0.37	0.34	0.38	0.44	0.41	0.55	0.56
	Salinas	OA	90.01±0.84	85.87±5.18	86.39±1.83	90.76±1.3	88.77±1.5	90.73±3.02	91.08±3.65
		AA	90.23±1.27	88.03±3.51	85.29±3.34	89.99±2.29	87.74±1.35	90.52±2.31	91.41±3.28
		K	0.89	0.84	0.85	0.9	0.87	0.90	0.9
	Houston	OA	53.72±3.37	53.68±6.7	51.2±2.86	62.42±2.15	53.78±4.57	73.35±6.93	73.64±3.41
		AA	50.02±3.7	49.31±6.66	48.34±3.84	58.84±1.6	50.37±4.87	70.86±6.85	71.3±4.01
		K	0.5	0.5	0.47	0.59	0.5	0.71	0.71

parameter and gamma in Online EWC are 50 and 1. In XDG, the percentage of hidden units to gate is set to 0.8. The regularization strength c in SI is 0.1. And other parameters, such as learning rate, are the same as our method, and the principles are described as follows.

EWC introduced fisher information to discriminate the important parameters of historical analyses, and it used penalty terms for all the previous task parameters to bring parameters of the new process close to already learned parameters. The computation overhead of maintaining penalty terms for the increasing number of tasks, however, grows as the number of tasks becomes larger. Compared with EWC, Online EWC only utilizes the most recent task parameters to constrain the new task updates when processing multiple tasks, avoiding the waste of computational resources. Similarly, the SI approach selectively reduces the learning rate of the importance weights of the tasks that need to be maintained. LWF applies a KD strategy to maintain the hidden knowledge from the historical model. This approach trains a new initialization model with new data while keeping the new data in the historical model output to construct distillation losses. XdG assumes that the executed task is already known, and randomly selected portions of the networks for each task are used to minimize disruptions between different tasks. In the end, we introduced a fine-tuning method without any memory mechanism to imitate the forgetting of the model.

Comparing with the above models, our proposed model achieves the best performance of continual classification tasks on the continual updated fourth hyperspectral datasets. As shown in Table VII, we further demonstrate the OA of the classification of the contained datasets in each task. In the continuous learning task applied to the four datasets, our proposed method improves the accuracy at least 15.39% and 19.92% compared to the EWC* method on Indian pines and Houston datasets at the step 4 task; however, these methods do not have continuous update datasets for new tasks. The XDG* is not able to learn quickly from new datasets, and only obtained 45.45% OA at step 2 when the addition of new Indian pines dataset.

In addition, when the continuous learning task is updated for all four datasets, these comparison models show that knowledge of the historical Pavia dataset was forgotten, increasing accuracy on the recent Indian pine dataset. Specifically, in the third step of the task, our proposed method improves at least 5.12%, 3.42%, and 5.71% on the Pavia dataset compared to representative EWC-online*, LWF*, and XDG* models. In summary, Table IV demonstrates that other comparison models keep a lower learning rate on the latest dataset while maintaining knowledge of the model learned earlier. Our proposed model achieved superior performance in learning new knowledge and maintaining historical knowledge.

As shown in Fig. 5, each point represents the AA of the corresponding model for each task. Since the accuracy results of these models are approximate in the beginning step, we display the average task accuracy in the following three steps tasks. Note that points are positioned closer to the upper right corner in the plot, indicating that the model has higher AA on these four tasks and that the model

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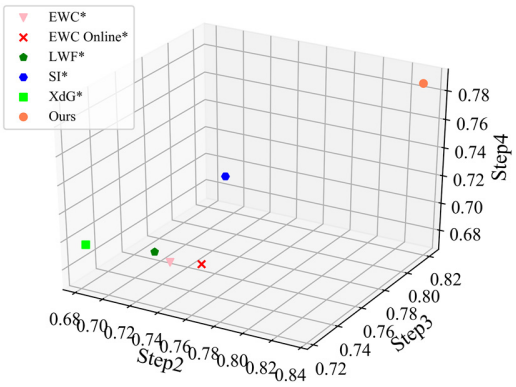

Fig. 5. AA results of different methods on continual learning tasks for step 2, step 3, and step 4.

performs better in the continuous learning process. The EWC*, EWC-online*, and SI* results are located close to each other in space due to their similar performance. Our model achieves higher performance in continuous learning over all four tasks stream. Specifically, Table VI shows the classification accuracy for each class for these datasets using the continuous learning process. Our model effectively retains the knowledge learned earlier by the model under continuous learning. Also, with the data retention mechanism, our proposed method shows only a slight decrease in the classification accuracy of each category on the beginning Pavia dataset. For the following four steps of the tasks stream, our method guarantees efficient learning of new knowledge from additional data and performances well at the second and third steps while the Indian pines and Salinas when the Indian pines dataset and the Salinas dataset are treated as fresh data in the streaming.

E. Visualization Results

We have demonstrated the quantitative classification accuracy table of our proposed method and the comparison methods for the continuous update tasks. In addition, we display the visualization results of all methods on the four-step tasks separately in Figs. 6–9. We present the first step classification results for the Pavia dataset for EWC*, EWC Online*, LWF*, SI*, XdG*, and our method in Fig. 6. Compared with the smooth and regular original ground truth, EWC*, EWC Online*, LWF*, SI* obtained coherent land cover results, but there is still amount of noise in the classification maps. It also appears the mix-up classification results across different datasets. Although there is less noisy data and small amount of cross-dataset confusion in the classification map of XdG*, the OA is worse.

On the second task, the noise in these classification maps gradually increases with the additional new Indian Pines dataset, especially the building area in the middle of the map. Meanwhile, the noise in the classification map persists through the third step. In contrast, our proposed method introduces a spatial-spectral loss function, which efficiently retains the spatial and spectral features of these HSI learned by the earlier model. As a result, our method achieved more accurate

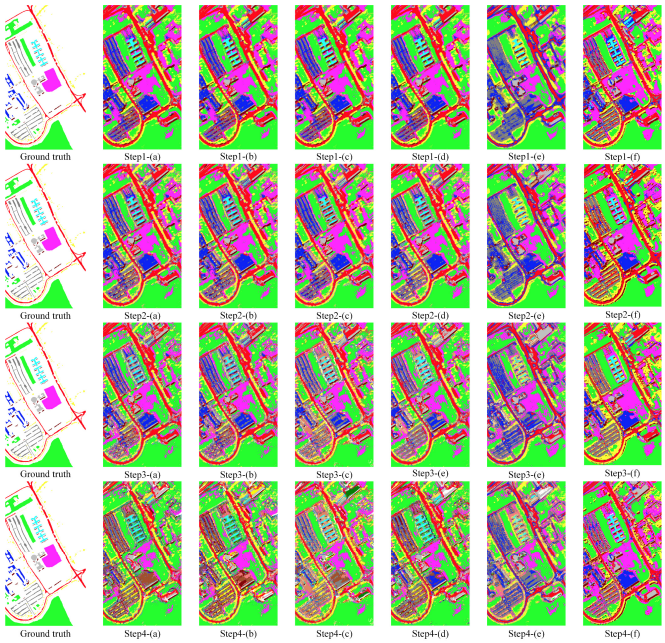

Fig. 6. Classification results of different methods on the Pavia dataset using 1% labeled samples for training. Ground truth represents visualized label, (a) EWC*, (b) EWC Online*, (c) LWF*, (d) SI*, (e) XdG*, (f) our method. Step 1, step 2, step 3, and step 4 represent four-step tasks, respectively.

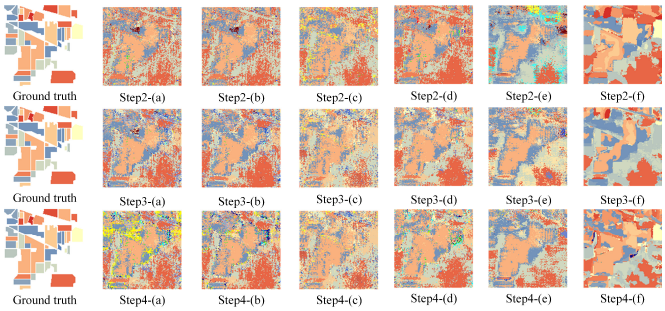

Fig. 7. Classification results of different methods on the Indian Pine dataset using 1% labeled samples for training. Ground truth represents visualized label, (a) EWC*, (b) EWC Online*, (c) LWF*, (d) SI*, (e) XdG*, (f) our method. Step 2, step 3, and step 4 represent second step, third step, and fourth step tasks, respectively.

labeling of the hyperspectral pixels, and our predicted classification map has a more regular shape and less noise after the continual learning process. In summary, although these models preserve as much of the learned knowledge as possible with the added data retention scheme, these comparison methods still failed to avoid the problem of catastrophic forgetting while adding new datasets for training.

Also, these models retain historical knowledge while failing to rapidly obtain new knowledge from new datasets. As shown in Fig. 7, we can find that in the second step task, there are massive misclassifications of the Indian Pines dataset seen in these comparisons. As for the LWF* and SI* methods, the figures indicate that it is difficult to learn new knowledge quickly on the Indian pines dataset, which results in poor performance. For EWC*, EWC online*, and XdG*, these methods failed to maintain the performance of the previous

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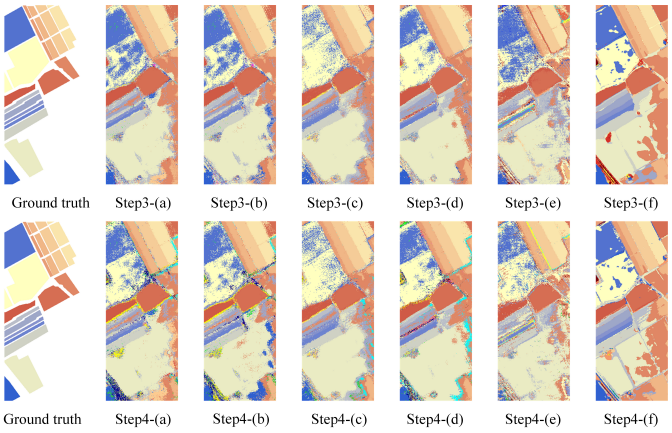

Fig. 8. Classification results of different methods on Salinas dataset using 1% labeled samples for training. Ground truth represents visualized label, (a) EWC*, (b) EWC Online*, (c) LWF*, (d) SI*, (e) XdG*, (f) our method. Step 3 and step 4 represents the third step and fourth step tasks.

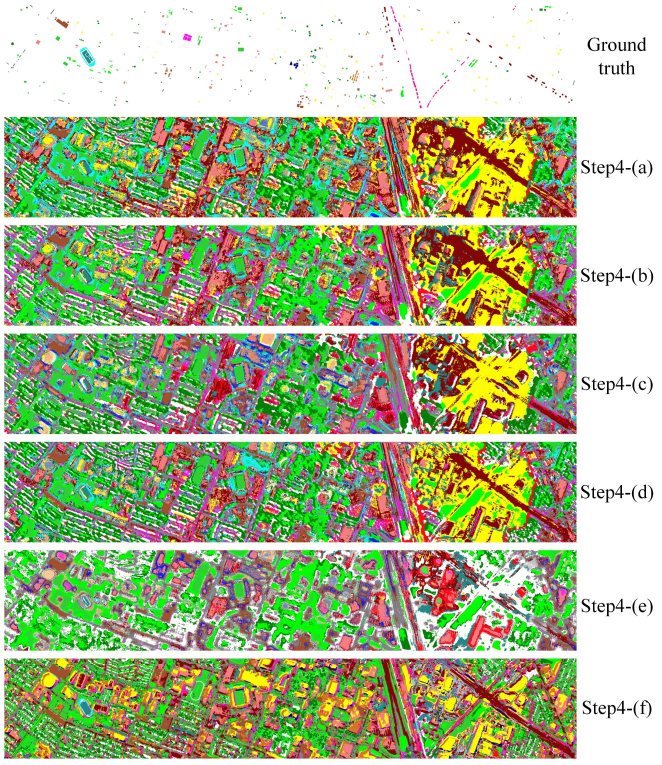

Fig. 9. Classification results of different methods on the Houston dataset using 1% labeled samples for training. Ground truth represents visualized label, (a) EWC*, (b) EWC Online*, (c) LWF*, (d) SI*, (e) XdG*, (f) our method. Step 4 represents the fourth step task.

model, and these methods are suffering from cross-domain confusion at different steps because of the different datasets. In summary, EWC*, Online EWC*, and XdG * fail to maintain the performance of the model on the Indian Pines dataset as the model is continuously updated. Our proposed method still achieved superior performance in the continual learning procedure. In particular, the designed spatial-spectral feature distillation mechanism helps the model obtained robust features and retain the learned knowledge effectively. Thus, our method obtained better accuracy and relatively smoother and regular results in the classification map at the final step task.

Similarly, in the third step, the performance of these models for the Salinas dataset also suffers from misclassifications across different datasets, which are shown in Fig. 8. There is a significant portion of misclassified noise data in the top area and appears cross-domain mix-up at the edge of different categories. Especially EWC* and SI* occur lots of cross-domain misidentified pixels in the boundary of different categories. And the accuracy of the new dataset fails to improve quickly when updating to the third task. Since our method effectively discriminates different domain data by incorporating spatial-spectral features, more regular and smoother results of classification are obtained. Although historical data can be used for training, it demonstrates that these methods would be influenced by other datasets and fail to keep historical knowledge and learning novel knowledge in balance. For example, in Fig. 9, with the addition of the latest Houston dataset, the classification results of these methods demonstrate some incorrect classification results, especially in the road area on the right side of the figure, which appears to merge with other categories. On the contrary, our proposed method performed well on the four-step continual learning tasks stream. It also enables learning new knowledge from new tasks very quickly while preventing catastrophic forgetting of the model.

V. Conclusion

In this study, we proposed a continual learning method for HSI classification across different datasets. To overcome the problem of catastrophic forgetting during the continual learning classification, we designed several loss functions to preserve the historical knowledge learned by the model. First, we employed the cross-entropy loss function as the fundamental loss function for continuous learning across all four HSI datasets. Then, the spatial feature loss and spectral feature loss are designed to retain the deep information of spatial and spectral features learned by the model. Finally, the class loss function is also introduced to encourage the high-level knowledge transfer to the new model. Several ablation experiments demonstrate the effectiveness of these loss functions, respectively. Experiments with other representative continual learning models have demonstrated that our proposed method achieved superior performance for continual HSI classification. In the future, continual learning under unlabeled conditions will be studied further. We will explore continual learning methods of labeled hyperspectral datasets to reduce the abundance of unlabeled data.

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